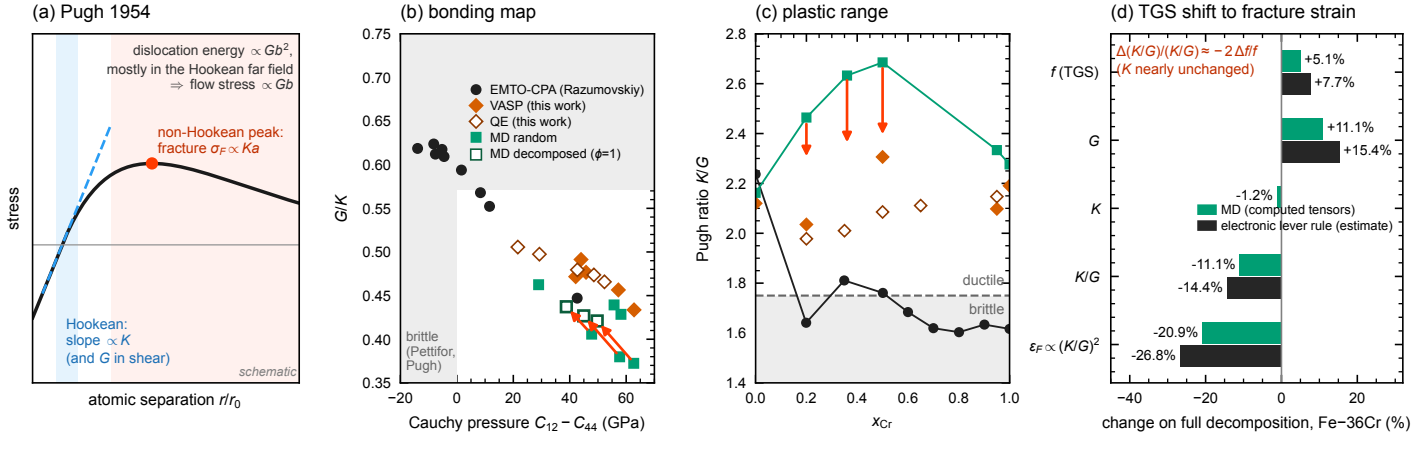
(a) Pugh 1954
dislocation energy ∝ Gb², mostly in the Hookean far field ⇒ flow stress ∝ Gb
non-Hookean peak: fracture σF ∝ Ka
Hookean: slope ∝ K (and G in shear)
schematic
stress
atomic separation r/r0
(b) bonding map
EMTO-CPA (Razumovskiy)
VASP (this work)
QE (this work)
MD random
MD decomposed (ϕ=1)
brittle (Pettifor, Pugh)
G/K
Cauchy pressure C12 − C44 (GPa)
(c) plastic range
Pugh ratio K/G
ductile
brittle
xCr
(d) TGS shift to fracture strain
Δ(K/G)/(K/G) ≈ −2Δf/f
(K nearly unchanged)
f (TGS)
G
K
K/G
εF ∝ (K/G)²
+5.1%
+7.7%
+11.1%
+15.4%
-1.2%
MD (computed tensors)
electronic lever rule (estimate)
-11.1%
-14.4%
-20.9%
-26.8%
change on full decomposition, Fe−36Cr (%)